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Large language models for newspaper sentiment analysis during COVID-19: The Guardian

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ABSTRACT

During the COVID-19 pandemic, news media coverage encompassed topics such as viral transmission, allocation of medical resources, and government response measures. Studies have analysed sentiment on social media platforms during COVID-19 to understand the public's response to the rising cases and the government strategies implemented to control the spread of the virus. Sentiment analysis can enable an understanding of the dynamics in the social and psychological well-being of a group during the pandemic. Apart from social media, newspapers have played a vital role in disseminating information, including information from the government, experts, and the public about various topics. A study of sentiment analysis of newspaper sources during COVID-19 for selected countries can give an overview of how the media covered the pandemic. In this study, we selected The Guardian newspaper and conducted a sentiment analysis during various stages of COVID-19 including initial transmission, lockdowns and vaccination. We employed novel large language models (LLMs) and refined them with expert-labelled sentiment analysis data. We also provide an analysis of sentiments experienced pre-pandemic for comparison. The results indicate that during the early pandemic stages, public sentiment prioritised urgent crisis response, later shifting focus to addressing the impact on health and the economy. In comparison with related studies about social media sentiment analyses, we found a discrepancy between The Guardian, with a dominance of negative sentiments (sad, annoyed, anxious and denial), suggesting that social media offered a more diverse emotional expression during the pandemic. We found a grim narrative in The Guardian with overall dominance of negative sentiments, pre- and during COVID-19 across news sections including Australia, UK, World News, and Opinion.

1. Introduction

The COVID-19 pandemic was a highly contagious [1] and catastrophic global event with infection symptoms such as fever and cough, with severe cases potentially leading to death [2]. Various measures were implemented to control the spread of COVID-19 such as patient isolation, mandatory masks in public places, social distancing, and virus testing and vaccination [3–5]. The pandemic led to social changes, including lockdowns, quarantines, economic downturns and loss of life, which have affected mental health [6]. Additionally, social uncertainty and fear brought about by the COVID-19 pandemic have led to social unrest and instability [7]. During the COVID-19 pandemic, major social media platforms such as Facebook, Weibo, and Twitter were widely used to disseminate information about infection cases [8]. They included government safety and control measures, vaccine and drug development, and various other aspects related to the pandemic [9]. Social media users shared their thoughts about the pandemic and expressed various concerns with different types of sentiments [10]. Social media has also been used as a medium for misinformation during COVID-19 [11] that resulted in measures by social media, such as the removal of fake accounts and better management of content [12].

Natural Language Processing (NLP) [13] is a field of artificial intelligence that enables computers to understand, interpret, manipulate, and generate human language [13]. The development of NLP has benefited from advances in machine learning and deep learning models [14–16], enabling NLP systems to yield favourable outcomes when tackling intricate language tasks [17]. Large Language Models (LLMs) refer to neural network models with a large number of parameters and extensive pre-training, primarily employed for natural language understanding tasks [18]. Sentiment analysis [19] in NLP aims to identify and understand the emotional tendencies and sentiments expressed within a text, typically by detecting sentiments such as positive, negative, and neutral [20]. Sentiment analysis has extensive applications across various domains, including social media analysis, public sentiment monitoring, product review analysis, and customer feedback analysis [19]. It has

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been demonstrated to be used not only for social media and advertising industry [19,21,22], but can also be used to study public behaviour in pandemic such as the COVID-19 [10] and compare texts to evaluate the quality of translations from low-resource languages [23].

Recurrent Neural Networks (RNNs) [24,25] are deep learning models for processing sequential data and are prominent for processing text data and performing tasks such as sentiment analysis. The Long Short-Term Memory (LSTM) network [26,27] is an advanced RNN for modelling data with long-term temporal dependencies. In the last decade, variants of the LSTM network have emerged, and the encoder–decoder LSTM [28] became a prominent variant for processing and modelling text data. The Transformer model is an advanced encoder–decoder LSTM with attention mechanism [29] that was designed for NLP tasks, particularly for tasks involving sequential data. It is currently the mainstream framework for building LLMs. The Transformer model lays the foundation of the BERT (Bidirectional Encoder Representations from Transformers) [30] model that is a pre-trained LLM used for various NLP tasks, including sentiment analysis [31]. RoBERTa (Robustly optimised BERT approach) [32] is a variant of BERT that utilises larger-scale datasets for pre-training, longer sequence lengths, and dynamic masking. These improvements enhanced the RoBERTa model performance and generalisation capability for sentiment analysis. BERT-based LLMs have been widely applied in sentiment analysis of content on social media. Wahyudi and Sibaroni [33] used BERT for sentiment analysis of TikTok reviews obtained from the Google Play Store and the Apple App Store. Kikkisetti et al. [34] utilised LLMs for identifying emerging coded antisemitic hate speech on extremist social media platforms.

During the COVID-19 pandemic, major news media outlets continued to track and report on the development of the epidemic [35], such as the US Daily newspaper [36], China Daily [37], and The Guardian [37]. Due to the large volume of news articles, researchers employed NLP methods to analyse news. Some of the prominent NLP applications for COVID-19 news are as follows. Kocaman et al. [38] used the clinical and biomedical NLP models from the *Spark NLP*[1] healthcare library to analyse COVID-19 news publications from the *CNN* and *Guardian* newspapers [39]. Evans and Jones [40] used NLP to analyse news from major UK news channels, aiming to explore the widespread negative impact of the COVID-19 pandemic on mental health. Through NLP, we can gain insights into the public opinion trends and the epidemic situation of COVID-19 across different countries. For example, Tejedor et al. [41] reported that Spain and Italy exhibited a high degree of politicisation in epidemic control, and Apuke et al. [42] reported that the news media in Nigeria focused more on reporting death tolls and typical cases rather than epidemic prevention measures, which led to public panic. These motivate further studies of newspaper reporting during COVID-19 via NLP methods such as sentiment analysis.

The Guardian is a prominent international newspaper that covers COVID-19; hence, we discuss some of the relevant studies in the analyses of its news reporting. Sheshadri et al. [43] conducted an analysis of selected news articles from The Guardian and The New York Times to investigate how certain hierarchies are expressed and evolve across time and place. The authors found that media portrayals of religious groups remained relatively consistent, showing stability across decades rather than being significantly influenced by immediate events. Tunca et al. [44] employed NLP methods to analyse news articles from The Guardian to identify sentiment characteristics and reveal the interrelationships between the concept of the metaverse [45] and other relevant concepts. The study reported that positive discourses about the metaverse were associated with key innovations for users and companies, and negative discourses were associated with issues relating to the use of social media platforms such as privacy, security and abuse.

Abbasian et al. [46] focused on media reflection during the Iraq War to examine the portrayal of British Prime Minister Tony Blair in The Guardian, and observed that Blair was portrayed as a war criminal by this newspaper.

Sentiment analysis has been used to identify mental health problems during COVID-19 for psychological intervention [47]. Sentiment analysis can be used to evaluate the effect of policy measures, since different policy measures may have different sentiments and reactions to the public. Goel et al. [48] assessed the impact of various policies on public sentiments, which provided feedback for government decision-making. Through sentiment analysis, we can understand the public's reaction to information during COVID-19, such as the COVID-19 vaccines [49]. By identifying positive and negative sentiments, we can determine the overall public response to vaccines and news events related to vaccines, to accurately convey information and guide the public to face the challenge with a positive attitude [49]. Therefore, sentiment analysis can help us better understand the public's emotional response before and during COVID-19.

In this study, we analyse the sentiments expressed in The Guardian newspaper before and during the COVID-19 pandemic. We investigate selected article sections and compare the difference in sentiment expressions between the World, UK and Australian article sections. We present a framework that employs pre-trained LLMs that include BERT and the RoBERTa models. The main goal of our study is to provide sentiment analysis prior to, and during, COVID-19. This can provide guidance for policymaking, crisis response, psychological intervention and public opinion guidance in planning for the management of critical events. We compare the sentiments obtained during COVID-19 with the trend of COVID-19 infections (death) and also provide analysis via a quarterly timeframe.

The rest of the study is organised as follows. In Section 2, we provide a background on deep learning models for LLMs. Section 3 presents the methodology, including data processing and modelling. In Section 4, we present and discuss the results and Section 5 provides a discussion of the major results and limitations. Finally, we conclude the paper in Section 6.

2. Background

2.1. Recurrent Neural Networks

Unlike traditional feedforward neural networks, RNNs feature recurrent connections, enabling them to capture temporal dependencies within sequential data [50,51]. RNNs have advantages in handling sequential data and capturing temporal dependencies with recurrent connections, thereby better understanding the contextual information, which makes them applicable to NLP [52]. Also, the recurrent structure enables the model to memorise information from previous time steps and propagate it into future steps. In terms of data input, RNNs have a certain degree of flexibility and can handle input sequences of various lengths. LSTM networks [27] provide an extension to conventional RNNs for modelling long-term dependencies in sequence data [26]. LSTM networks address the issue of vanishing gradients [53] encountered by conventional RNNs. LSTMs demand significant computational power and memory, especially when training on large datasets [54]. Additionally, LSTMs are susceptible to overfitting, particularly in scenarios with relatively small data volumes, although this issue can be mitigated through regularisation methods such as dropouts [55]. Despite these limitations, due to their ability to capture long-term dependencies and generalisation performance, LSTMs remain an effective tool for NLP.

2.2. Transformer model

The Transformer model is an advanced RNN effective in processing sequential data, such as text or time series information [56]. The

[1] https://sparknlp.org/

Table 1
The Guardian's main sections and article counts.

Section name	Article count
World News	11 269
Opinion	7912
Football	7135
Sport	6633
Australia News	5419
US News	5004
Business	4672
Politics	4606
UK News	4255

Transformer model utilises an *attention* mechanism and utilises the bidirectional LSTM network [57]. The attentional mechanism is crucial for understanding the key points in a text segment and provides better contextual understanding. Training on large datasets also aids the model in understanding rare words or expressions [58]. Additionally, the Transformer model can be fine-tuned for specific tasks and the accuracy can be improved based on large-scale datasets for specific tasks [59]. However, particularly large Transformer models require significant computational resources for training, including considerable time and memory resources, which makes the cost of maintaining these models quite high [60]. Furthermore, fine-tuning Transformer models on small datasets may lead to overfitting, where the model excessively learns specific patterns in the training data at the expense of its generalisation ability [61].

2.3. BERT and RoBERTa models

The BERT is a pre-trained LLM based on the Transformer model [30] and capable of understanding the bidirectional context of each word within the text. This is distinctly different from conventional unidirectional or fixed-direction language models. BERT operates in two main phases: pre-training and fine-tuning. During pre-training, it engages in two types of self-supervised learning: masked language modelling (MLM), which involves predicting tokens for randomly masked positions in the input, and next sentence prediction (NSP), which determines if two segments of text are sequentially related. For fine-tuning, which tailors the model for specific tasks, additional fully connected layers are appended above the final encoder output. The inputs can be prepared by tokenising text into subwords, and then merging three types of embeddings – token, positional, and segment – to form a standardised vector.

The RoBERTa model [32] is a pre-trained LLM that adopts the "robust optimisation BERT method" in its design. In comparison to the original BERT model, RoBERTa shows several advantages and drawbacks. RoBERTa benefits from a larger training corpus, containing extensive text data from web pages, forums, books, and more, thereby enhancing the model's performance. The model employs a stricter masking strategy (dynamic masking) by replacing all words in the input text with "[MASK]", facilitating more effective utilisation of training data information. Furthermore, RoBERTa adopts a deeper network architecture, which contributes to further improving the model's performance. Finally, it demonstrates superior performance across various NLP tasks, including the GLUE (General Language Understanding Evaluation) [62] and SuperGLUE [63] task sets. However, there are drawbacks to consider since RoBERTa requires a larger volume of text data and a deeper network structure in comparison to BERT, resulting in longer training times. The substantial model size of RoBERTa poses challenges in terms of resource requirements during deployment.

3. Methodology

3.1. Data

3.1.1. SenWave sentiment dataset

The SenWave dataset [64] features over 105 million tweets and Weibo messages on COVID-19 from March 1 to May 15 2020, in six distinct languages: English, Spanish, French, Arabic, Italian, and Chinese. This dataset features 10,000 English and 10,000 Arabic tweets that have been tagged across 10 specific sentiment categories. Additionally, it includes 21,173 Weibo posts categorised into 7 different sentiment types for the Mandarin language. In total, the dataset contains 41,000 labelled items and over 105 million unlabelled tweets and Weibo posts, positioning it as the most extensive labelled COVID-19 sentiment analysis dataset. In our study, we utilise a subset of the SenWave dataset,[2] specifically focusing on 10,000 English tweets with sentiment labels such as optimistic, thankful, empathetic, pessimistic, anxious, sad, annoyed, denial, official COVID-19 report, and joking.

The first phase of our framework focuses on the preprocessing of the SenWave data using NLP tools. This includes converting text to lowercase, expanding contractions, translating emojis to text, and processing special characters and URLs using regular expressions and custom functions as done by Chandra and Krishna [10]. This process ensures that the data is homogenised for further analysis. We train our model using the SenWave dataset, and then apply sentiment analysis on The Guardian newspaper articles covering COVID-19.

3.1.2. The Guardian newspaper

The Guardian is a well-known British news media that covers local and international news. *Kaggle*[3] is a popular platform for the data science community, where people can find a rich variety of datasets covering fields such as finance, healthcare, image processing, and NLP. Kaggle has been prominent in running forecasting and machine learning competitions [65]. The Guardian newspaper dataset on Kaggle [66] comprises approximately 150,000 news articles from 2016 to 2023. We extracted and partitioned the dataset into two main intervals based on the World Health Organization's (WHO) COVID-19 timeline [67]: pre-pandemic (January 1, 2018, to December 31, 2019) and during the pandemic (January 1, 2020, to March 31, 2022). We focused on quarterly (3 months) analysis of the data and divided the dataset into nine distinct periods for observing the evolution of media sentiments as the pandemic unfolded. This is also to identify shifts in focus and tone within the news, revealing how the crisis influenced reporting priorities and language use over time.

The Guardian dataset is categorised into 164 sections, and a significant number of these sections contain only one or two articles, hence, we excluded these sections from further analysis. We selected only four major news sections for detailed examination, including World News, Opinion, Australia News, and UK News. We omitted both the Football and Sports sections, despite their substantial data (combined volume exceeding 10,000 articles, as shown in Table 1). The rationale behind this exclusion lies in the assumption that sports-related content would unlikely contribute meaningful insights relevant to our study.

Our study focuses on sentiment trends between the UK and Australian news, acknowledging a discrepancy in the volume of articles, with the Australian dataset surpassing the UK's by approximately 1,000 articles. Additionally, the Opinion section was specifically selected for its unique editorial stance. Unlike other sections aiming for neutrality, the Opinion section inherently encompasses a broader spectrum of sentiments and viewpoints, providing a rich source for dissecting sentiment shifts before and amidst the pandemic. This choice stems from the premise that newspaper opinion pieces are more likely to express sentiments that deviate from neutrality. However, they can offer invaluable insights into the public and editorial sentiment during these contrasting periods.

[2] SenWave dataset: https://github.com/gitdevqiang/SenWave/blob/main/labeledtweets/labeledEn.csv.

[3] https://www.kaggle.com/

3.2. N-gram analysis

N-gram is a commonly used for NLP that reviews contiguous N-elements (usually words or characters) in text data [68]. The N-gram model assumes that each element in the sequence depends only on its previous element, and utilises this local dependency to analyse data. The advantages of the N-gram model include simplicity, computational efficiency, and its ability to capture local information [69]. However, the N-gram model cannot capture long-distance dependencies and faces challenges in handling rare combinations. Lyse and Andersen [70] used N-gram in news media analysis to rank word sequences from the Norwegian Newspaper Corpus to assess the propensity co-occurrence of binary and triplets. Furthermore, N-gram analysis was used for selected Persian [71], German [72], and English newspapers [73]. Therefore in our study, we use N-gram analysis to complement sentiment analysis by LLMs.

3.3. Fine-tuning BERT-based models

Given the diverse range of topics and writing styles found in our news articles dataset, fine-tuning the BERT and RoBERTa models is crucial so that they can effectively provide sentiment analysis using the SenWave training data [64]. However, fine-tuning BERT requires setting up a suitable environment to enhance efficiency and effectiveness [74]. Furthermore, news articles often present lengthy content, which will surpass BERT's token limit. Segmenting these articles and aggregating sentiment scores poses another challenge, necessitating careful attention to maintain context and coherence [75]. Moreover, the presence of sarcasm and nuanced sentiments in news articles can perplex the model and hinder interpretation. Therefore, handling specialised terminology, proper nouns, and novel words absent from BERT's vocabulary is important for better sentiment analysis [76]. Addressing these challenges needs a multifaceted approach involving meticulous data preprocessing, domain-specific fine-tuning, and appropriate parameter setting. These measures collectively ensure the accurate and reliable sentiment analysis of news articles using BERT.

3.4. Framework

Our framework for sentiment analysis of newspaper articles includes several consecutive steps, as shown in Fig. 1. The major components include: 1. data selection; 2. data cleaning; 3. bigram and trigram visualisation; 4. fine-tuning BERT and RoBERTa using SenWave data; 5. newspaper article-level sentiment analysis ; 6. multi-label sentiment classification; 7. visualisation and comparison.

Our framework uses two data sets, including the Guardian articles and the SenWave dataset. We use the SenWave dataset for refining the LLM models. We first obtain the Guardian dataset from Kaggle and filter it based on publication dates and section names. We extract articles tagged (sections) as Australian news, UK news, World news, Business, Politics and Opinion. Note that the Opinion section encompasses local (UK) and international contributors. We believe that these sections are more important in sentiment analysis during COVID-19 since we cover the country of source of the Guardian, the world, and Australia which is geographically far but politically aligned with Great Britain.

In data processing (Step 2), we remove the items such as hashtags, stop words and web hyperlinks from the Guardian data using NLTK (Natural Language Toolkit) [77] tools. In order to obtain more meaningful bigrams and trigrams, and subsequent sentiment analysis, we also remove stop words based on categories, such as general terms, as shown in Table 2. We extend our data cleaning process to the SenWave dataset, which consists of tweet texts, unlike the formal media articles from The Guardian. Given the dynamic nature of social language expressions, we account for abbreviations, emoticons, and other informal elements as outlined by Chandra and Krishna [10] for processing the SenWave dataset for model training (refinement) and testing. It is worth noting that within the SenWave dataset, the term "official report", while typically considered a topic, is classified as a sentiment. This distinction is noteworthy, especially considering that in our analysis of The Guardian articles, sections discussing "official reports" comprise a significant portion of the dataset.

In Step 3, beginning from the year 2018, we generate all possible bigrams and trigrams for each quarter (data spanning every three months). We then visualise the bigrams and trigrams and analyse word combinations and frequencies to gain insight into article characteristics across different periods (quarters) covering COVID-19.

In Step 4, we leverage pre-trained BERT and RoBERTa models via Hugging Face[4] which is a company dedicated to providing open-source NLP tools [78]. One of their flagship products is the Transformer model library, which provides a variety of pre-trained deep learning models, including BERT and RoBERTa. Hence, we can access these state-of-the-art models and fine-tune them for our sentiment analysis tasks to ensure optimal preparation. We employ the cleaned SenWave dataset for refining the pre-trained models (BERT and RoBERTa) using GLoVe embedding [79] for multi-label sentiment classification. We note that BERT and RoBERTa feature word embedding, but we used GloVe based on earlier work that also utilised BERT with SenWave date for refinement [10].

We begin by setting parameters, including maximum token length, batch sizes, epochs, and the learning rate to tailor the training process. Using the BERT and RoBERTa Tokenizers from the Hugging Face Transformers library [80], we convert the tweets into a compatible format for processing. We enhance the model with a dropout layer for regularisation and a linear layer for the output. Model training involves iterating over batches, where the model computes losses and updates weights accordingly. After training, we evaluate the model's performance using metrics such as Hamming loss and Jaccard scores. These metrics provide critical insights into the model's effectiveness in handling multi-label classification tasks, underscoring our comprehensive approach from data preparation to model training and evaluation in a deep learning framework.

In Step 5, we utilise the fine-tuned BERT and RoBERTa models to conduct sentiment analysis on selected articles from the Guardian. In Step 6, we conduct a multi-label sentiment classification of the Guardian data across the different quarters by categorising sentiments such as optimism, anxiety, pessimism, and surprise. This approach allows for a more nuanced understanding of the emotional nuances expressed within the articles during different phases of COVID-19 that spans quarters before and during the COVID-19 pandemic, covering the beginning, lockdowns, vaccination, end of lockdowns and unrestricted travel, locally and internationally, which marks the end of COVID-19.

In the final step (Step 7), we employ various visualisation techniques, including bigram and trigram analysis for selected sentiments, sentiment polarity, sentiment distribution histograms, and heatmaps. We compare the different sections of The Guardian before and during the COVID-19 pandemic. Through these visualisations, we aim to illustrate the evolution of sentiments and topics over time.

3.4.1. Technical details

Our framework in Fig. 1 leveraged the BERT-based pre-trained models from Hugging Face [80]. We implemented our framework in Python with libraries such as NumPy and Pandas for data manipulation, Matplotlib and Seaborn for data visualisation, and NLTK[5] with TextBlob[6] for NLP tasks. We also used PyTorch library that supports vector and tensor operations and model training processes, alongside TorchText[7] library for handling text data. We provide open source code

[4] Hugging Face: https://huggingface.co/models

[5] https://www.nltk.org/

[6] https://textblob.readthedocs.io/en/dev/

[7] https://pytorch.org/text/stable/index.html

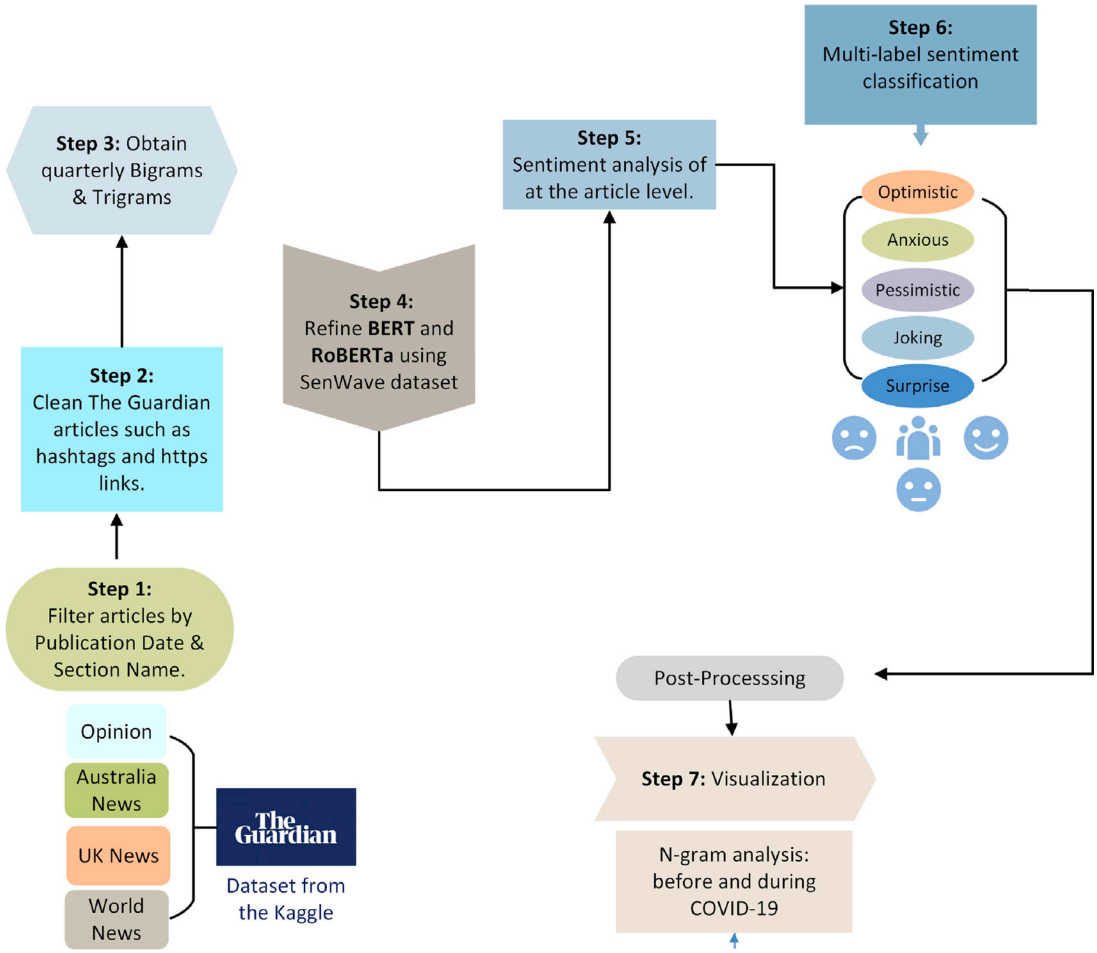


Fig. 1. A sentiment analysis framework for The Guardian articles (01/01/2018–31/03/2022) utilising BERT and RoBERTa models trained and tested on the SenWave dataset.

Table 2
Category of extra stop words used for the N-gram analysis. These stop words are not part of the default English dictionary of stop words obtained from NLTK[a].

Category	Stop words
General terms	guardian, article, theguardian, com, says, also, world, news, report, way, ways, use, used, using, place, places
People and entities	man, men, woman, women, child, children, company, companies, minster, government, scott, morrison, police, officer, boris, johnson, federal, prime, york, zealand
Action verbs	make, makes, made, go, going, get, getting, got, see, seeing, seen, take, takes, took, come, comes, came, look, looks, looking, help, want, wants, wanted, tell, tells, told, work, works, worked
Time-related	year, years, time, times, week, weeks, month, months, day, days
Quantifiers and qualifiers	one, many, much, lot, lots, number, numbers, group, groups, good, bad, important, need, needs, needed, including
Modifiers and auxiliaries	said, just, like, can, may, might, think, thinks, thought, should, must, do, does, will, would, could, south, wale

[a] https://www.nltk.org/search.html?q=stopwords

and data in our GitHub repository, with details at the end of the paper.

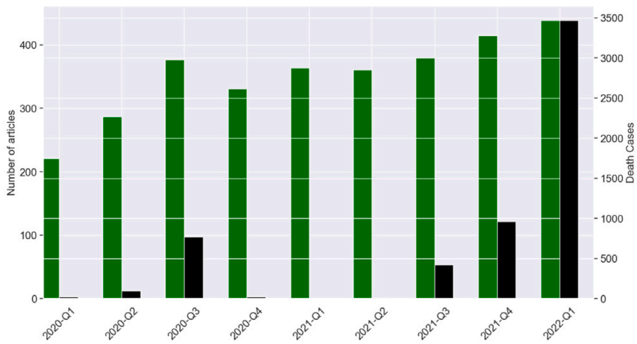
We fine-tuned (refined) both models (BERT and RoBERTa) to approximately 400 megabytes in size using SenWave dataset for sentiment analysis. We used the BERT-base-uncased version, which contains approximately 110 million parameters, and RoBERTa-base with 120 million parameters. We employed the BertTokenizer to prepare the input data with a maximum sequence length of 200 tokens to align with the average length of tweets.

We used a computer with 1 Intel Core i7-12700H processor and 16 GB of memory. In order to validate the SenWave-based refinement of the model, we implemented a 90:10 ratio for training and test sets. We employed a relatively low learning rate of 1e–05 for BERT and 2e–05 for RoBERTa, to achieve finer adjustments in the model weights during backpropagation. The training (refinement) process spanned 4 epochs and throughout the training process, the loss of the models showed a decreasing trend, reflecting continuous learning and improvement.

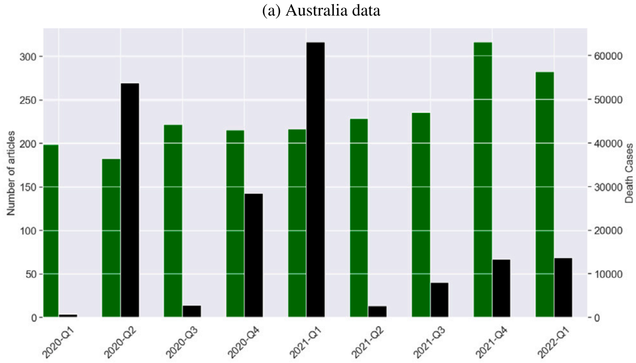
4. Results

4.1. COVID-19 - cases and deaths

We first present statistics about COVID-19 deaths in relation to the Australia and UK news sections of The Guardian, respectively. Fig. 2 describes a comparative analysis of The Guardian article counts and number of death cases across nine quarters for the given years (from the first quarter of 2020 to the first quarter of 2022). We extracted the COVID-19 death data from the Australian Bureau of Statistics [81] and the UK's Office for National Statistics [82]. In the case of Australia (Fig. 2 - Panel (a)), we notice the spikes in the COVID-19 death counts

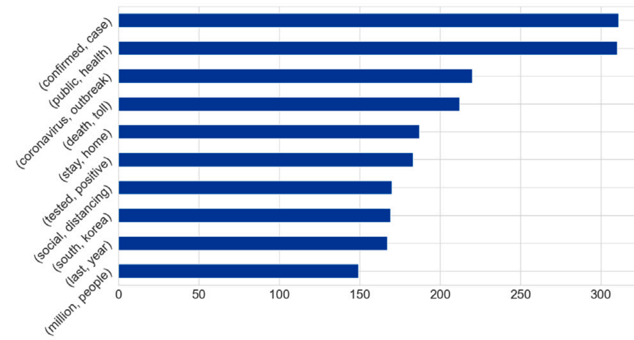


(a) Australia data

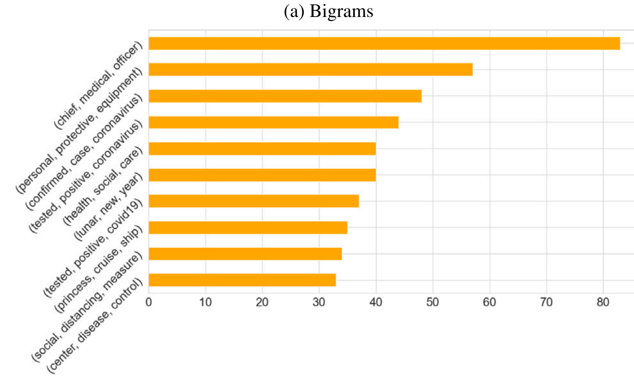


(b) UK data

Fig. 2. Comparative analysis of the number of articles (green) and death cases (black) in Australia and UK news sections of The Guardian.



(a) Bigrams



(b) Trigrams

Fig. 3. Top 10 bigrams and trigrams for World News in 2020 first quarter (January–March) covering the beginning of COVID-19 pandemic.

in the third quarter of 2020, the fourth quarter of 2021, and the first quarter of 2022. These spikes correspond to an increase in the volume of news content (articles), suggesting a correlation between the pandemic's severity and media coverage. In contrast, the UK experienced significant rises in death rates in the second quarter of 2020 and the first quarter of 2021, without an increase in the news report, indicating a possible divergence between public health data and media response. This analysis sets the stage for a focused examination of the specified quarters in subsequent, more detailed quarterly analyses. Furthermore, we also note that both countries have a major difference in population and the UK had three times more death per capita count when compared to Australia [83].

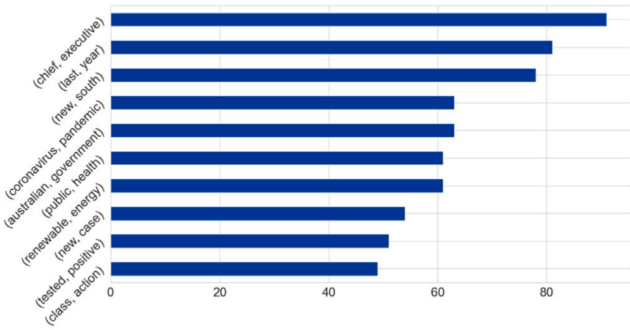
4.2. Data analysis using n-grams

Next, we use n-gram analysis to get an overview of the major topics expressed in the selected sections of The Guardian during the first phase of COVID-19. According to Fig. 3-Panel (a), in the first quarter of 2020, the lexicon of global news was dominated by phrases deeply tied to the emerging COVID-19 pandemic, with "public health", "confirmed case", and "coronavirus outbreak" leading the frequency plots in bigrams analysis. The trigrams (Fig. 3-Panel (b)) show "chief medical officer", "public health England", and "confirmed case coronavirus" emerged as top phrases in the World News section. The phrase "confirmed case" in both bigrams and trigrams points to the critical importance of tracking the virus at the beginning of the pandemic. Similarly, the frequent appearance of "personal protective equipment" in discussions signals the pivotal role of protective gear in controlling the outbreak, hinting at the global surge in demand for such equipment and the supply chain challenges that ensued. This lexical analysis of early 2020 news coverage offers a window into the world's collective focus and concerns during the initial stages of the COVID-19 pandemic, revealing an acute awareness of the crisis's impact on public health.

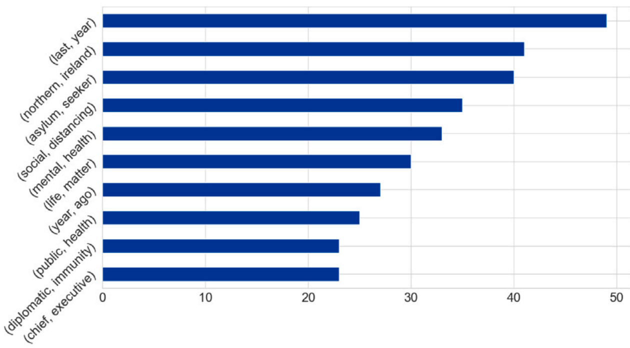
In the second quarter of 2020 (Fig. 4), analysis of bigrams in Australian and UK news highlights key concerns related to the COVID-19 pandemic. We find that Australia focused on "coronavirus pandemic", "public health", "new case", and "tested positive", reflecting close attention to the outbreak's evolution and its health implications. In contrast, the UK emphasised "social distancing", "mental health", and "public health", pointing to a broader scope of concerns that included preventive measures and the psychological impact of the pandemic. This contrast reveals differing priorities, where Australia's media concentrated on tracking and testing the virus, whereas the UK highlighted the social and mental health dimensions of the crisis. Through this lens, we can review how each country's media spotlighted shared and distinct aspects of the pandemic, underscoring diverse strategies and responses to the pandemic.

4.3. Fine-tuning BERT-based models

We present the training (model refinement) performance metrics for both the BERT and RoBERTa models using the SenWave training dataset with 90:10 train and test data split. The given metrics encompass Hamming loss, Jaccard score, label ranking average precision (LRAP) score, macro and micro F1 scores which are key performance metrics for multi-label classification in the literature [10,84]. In the Hamming loss, lower values signify better performance and the Jaccard score assesses the similarity between predicted and true labels, where higher values indicate better performance.The LRAP score evaluates the ranking quality of predictions based on the average precision of true labels, with higher values indicating better ranking performance. The F1 macro-score computes the average F1 score across all classes, reflecting precision and recall equally. Similarly, the F1 micro-score considers the overall F1 score, useful for addressing class imbalance. The higher values of the respective F1 scores signify better model performance. We present the results in Table 3, where RoBERTa exhibits marginally



(a) Australia news



(b) UK news

Fig. 4. Top 10 bigrams for Australia news and UK news in 2020 s quarter (April–June) covering the beginning of lockdowns of the COVID-19 pandemic.

Table 3
Refining BERT and RoBERTa models using SenWave test dataset.

Metric	BERT	RoBERTa
Hamming loss	0.147	0.135
Jaccard score	0.496	0.519
LRAP score	0.753	0.774
F1 Macro-score	0.536	0.533
F1 Micro-score	0.576	0.591

superior performance across LRAP, Jaccard and F1 micro scores. This indicates a slightly better overall performance when compared to the BERT model, which was also reported in previous work [10].

4.4. Sentiments detected by LLMs

We next compare the sentiments detected by BERT and RoBERTa models. Fig. 5 presents sentiment distribution within The Guardian (Australia, UK, World news and Opinion sections), spanning from January 1, 2018, to March 31, 2022.

A significant finding from Fig. 5 is the prevalence of "official report" as the dominant sentiment across the dataset, regardless of the model. This observation aligns with the nature of news reporting, which is fundamentally centred on neutrality and objectivity. Certain news articles conveyed information from official statements and reports during COVID-19. Therefore, the prominence of "official report" as a sentiment category underscores the adherence of the news media to these journalistic standards. Although the prevalence of the "official report" category is expected given the context of news reporting, it presents a challenge for our analysis focused on capturing a broader spectrum of sentiments related to the COVID-19 pandemic. This category's over-representation may obscure the nuances of public sentiment and emotional responses to the pandemic, as reflected in news coverage. Consequently, to address this issue and enhance the clarity of human sentiment expressed, we excluded the "official report" from

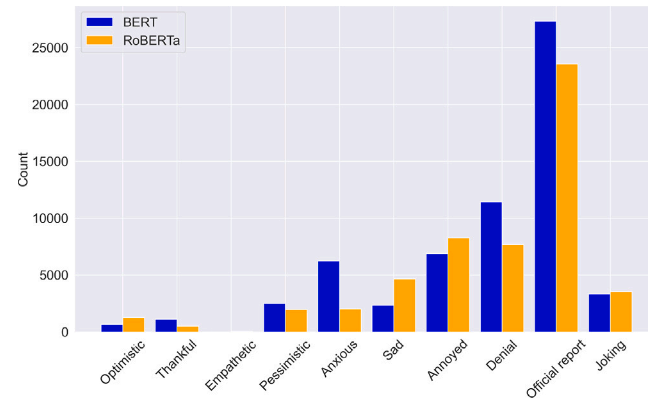


Fig. 5. Sentiment distribution by the BERT and RoBERTa models from The Guardian (Australia, UK, World news and Opinion sections combined), spanning from 1st January, 2018–31st March, 2022.

subsequent analyses. This will allow for a more nuanced exploration of the emotional landscape of news coverage during the specified period. We further uncover distinct emotional patterns and shifts that may provide deeper insights into the collective sentiment towards the pandemic, beyond the confines of official reporting. Finally, we notice that the "official report", "annoyed", and "denial" make the major differences between the sentiment counts when comparing BERT with RoBERTa in Fig. 5.

Now that we have compared BERT with RoBERTa, we will present the rest of the analyses using RoBERTa as our designated LLM. Fig. 6 displays the distribution of sentiments within Australian, UK, and World news sections, providing an intriguing perspective on how location-specific contexts influence sentiment portrayal. Despite the smaller dataset for the UK, with a thousand fewer articles than in Australia, an interesting pattern emerges. The sentiment "sad" is more pronounced in UK coverage, which reflects the higher COVID-19 mortality rates experienced in the UK when compared to Australia during the pandemic [83]. Moreover, it is noteworthy that sentiments such as "optimism", "thankful", and "joking" are more prevalent in the UK news when compared to Australia. Both the UK and Australia have reported a much lower number of negative sentiments (sad, annoyed, anxious, and denial) when compared to the World News section. This phenomenon might indicate a cultural or editorial leaning towards maintaining a semblance of hope and gratitude, even when faced with dire circumstances. The higher occurrence of these positive emotions, despite the adversity, aligns with our previous observations of an increase in such sentiments during the pandemic period. Furthermore, it could also indicate a political bias of The Guardian in reporting negative news for the rest of the world and more positive news about the Western world (UK and Australia).

In Fig. 7, we filtered articles from the Australia news section spanning January 1, 2018–December 31, 2019. We selected data for this trigram based on the three sentiments with the highest proportion ("sad", "annoyed", and "denial") in 2018 and 2019, as shown in Fig. A.19. In Fig. 7, we observe that the trigrams encompass key topics, including economics, politics, and military affairs.

4.5. Sentiment polarity analysis

In the global impact of COVID-19, media reporting on pandemic-related information was influenced by various factors, including government response measures, and public opinion. Consequently, media coverage would exhibit varying sentiments depending on the nature and subject of the news coverage. Analysing changes in sentiment polarity in Guardian articles before and during COVID-19 can provide insight into the major sentiments expressed in the different phases. We

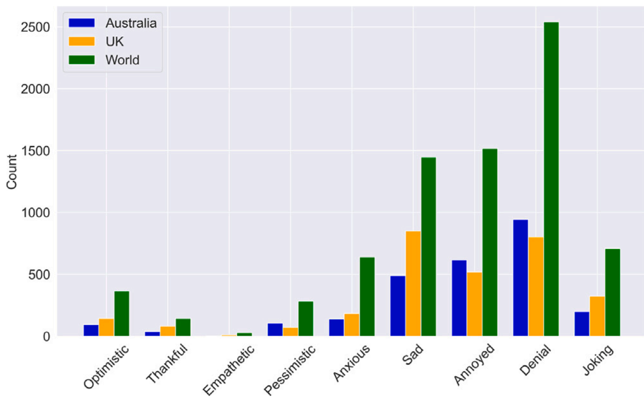


Fig. 6. Sentiments detected in Australia, UK and World news sections cover 1st January, 2018–31st March, 2022.

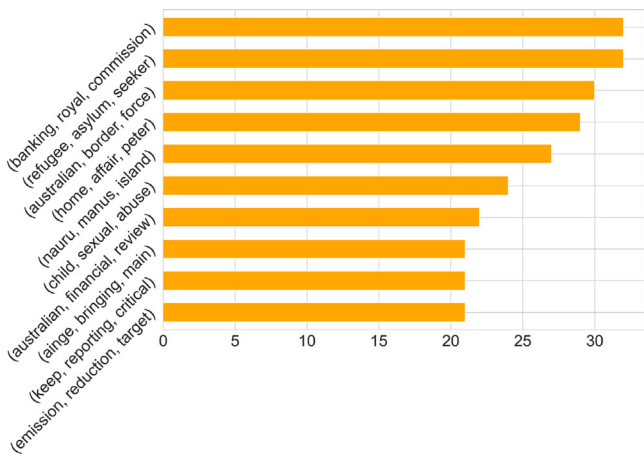


Fig. 7. Top 10 trigrams of the Australia news before COVID-19 (2018–2019).

Table 4
Sentiments and weights adapted from the literature [85].

Sentiment	Weight
Optimistic	1
Thankful	0.5
Empathetic	0
Pessimistic	−1
Anxious	−0.75
Sad	−0.75
Annoyed	−0.5
Denial	−1
Official report	0
Joking	0.5

assigned weights to each sentiment to compute our custom sentiment polarity scores for the Guardian articles using RoBERTa, as presented in Table 4. These scores range from −1 to 1, where a positive score denotes favourable sentiment towards the statement, and a negative score suggests adverse sentiment.

Sentiment polarity score in these sections of The Guardian increased significantly from 2019 to 2020, at the start of COVID-19, as shown in Fig. 8. According to an article published in The Guardian with the title "The Guardian in 2020" [86], this surge can be attributed to increased empathy, positive media narratives and people's ability to adapt. There are articles about solidarity, hope, and inspiration despite the challenges of COVID-19, creating an upward and positive sentiment trend.

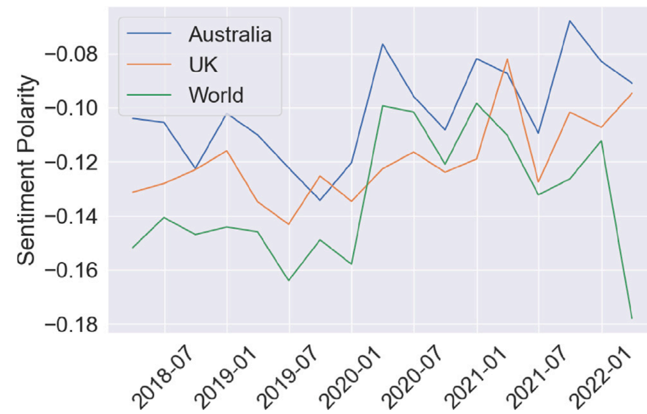


Fig. 8. Selected regions' quarterly polarity score fluctuation.

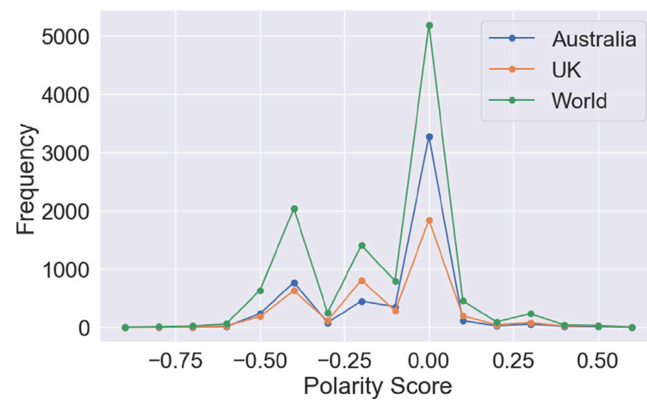
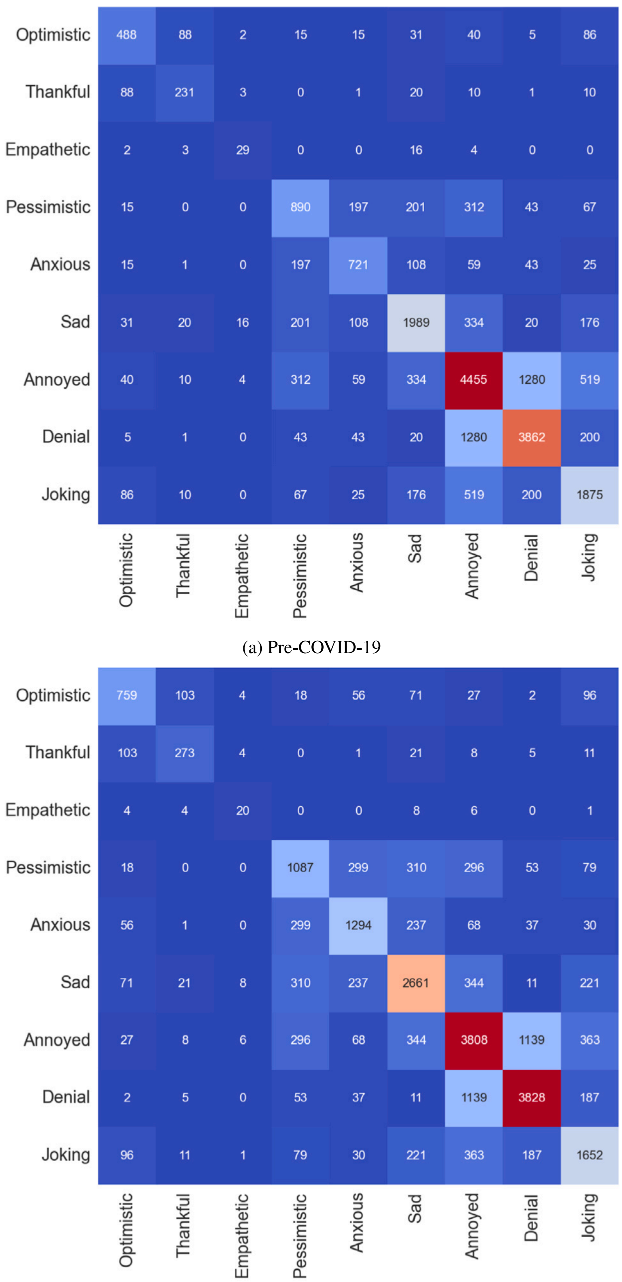


Fig. 9. Distribution of polarity scores for different regions for COVID-19 pandemic from 1st Jan 2018 to 31st March 2022.

According to Fig. 8, in the overall analysis, the polarity of pre-COVID-19 and post-COVID-19 articles tends to lean towards negative sentiment, which may be attributed to the higher proportion of negative sentiment in the weights we assign. We focus on data from the first quarter (January–March) 2018 to the first quarter of 2022, including Australia and the UK sections. During this period, articles in all three sections of The Guardian showed swings in sentiment polarity, reflecting the emotional ups and downs experienced by individuals during the COVID-19 pandemic. As shown in Fig. 9, the sentiment polarity of The Guardian is heavily skewed towards the negative, indicating the prevalence of emotional distress among individuals in the COVID-19 era.

Fig. 8 presents sentiment polarity in the Australian section of The Guardian, where we observe a downward trend from the second quarter of 2020 to the fourth quarter of 2020, possibly due to the rapid increase in global epidemic cases during this period. The sentiment polarity then continued to move downward towards the last quarter of 2021 which may be related to Australia's surge in new cases and socioeconomic conditions over the same period. Furthermore, the rebound in sentiment polarity in the UK in the first quarter of 2022 may be attributed to the stabilisation of the pandemic in the country, as shown in Fig. 2. The overall trend in the global sentiment polarity fluctuated downward throughout the COVID-19 period, particularly evident in the first quarter of 2022, suggesting a potential negative impact of the epidemic on global emotional well-being. The regional differences in sentiment polarity values may be associated with factors such as epidemic response measures, changes in vaccination rates, and economic recovery efforts in each region.



(a) Pre-COVID-19

(b) During COVID-19

Fig. 10. Heatmap showing the relationship between different sentiments (a) before and (b) during COVID-19 using the RoBERTa model. Note the data features Australia, UK, World News and Opinion sections.

4.6. Sentiments differences pre-COVID-19 and during COVID-19

We need to review the impact of COVID-19 on the style of the news coverage by The Guardian, and this can be done by comparing the sentiment analysis results pre-COVID-19 and during COVID-19. Figs. 10 and 11 present sentiment distribution and correlation using the RoBERTa model, spanning across the pre-pandemic and pandemic periods. We can observe that both periods are predominantly characterised by negative sentiments, where "denial", "annoyed", and "anxious" are the most frequently occurring sentiments detected by the BERT model. Conversely, RoBERTa identified "annoyed" as the most common sentiment, followed closely by "denial" and "sad". The onset of the pandemic brought about a noticeable increase in "anxious" and "sad" sentiment labels across both models, reflecting the global uncertainty and grief triggered by the crisis. This shift underscores the deep emotional impact the pandemic has had on societal sentiments, resonating through the tone of news coverage. Despite the surge in negative sentiments, the models also registered an uptick in positive sentiments such as "optimism" and "thankful" during the pandemic.

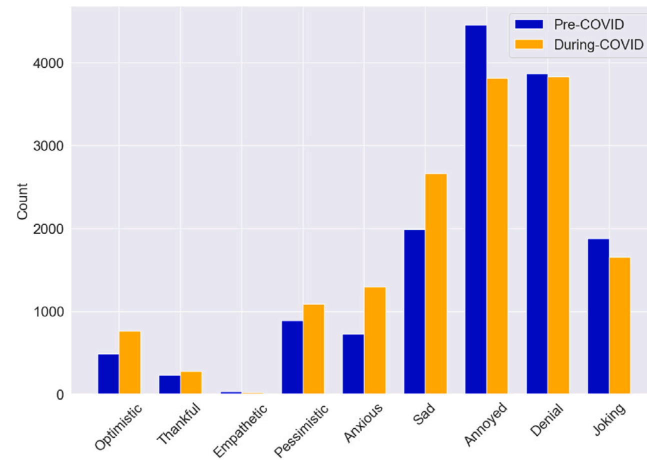
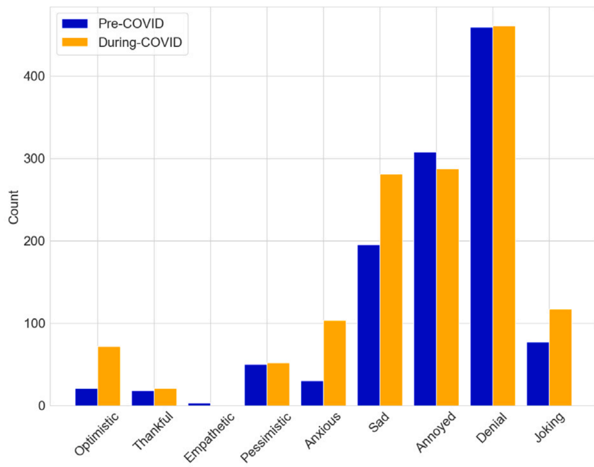


Fig. 11. Sentiments detected before and during COVID-19 using the RoBERTa model with data from Australia, UK, World News, and Opinion sections combined.

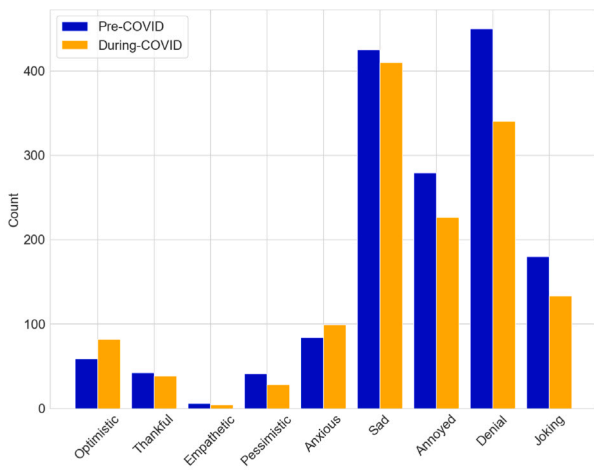
In Fig. 12, we can easily see that negative sentiments such as "denial", "sad" and "annoyed" are the most prominent in Australia, UK and World News sections, before and during COVID-19. Comparing the sentimental changes before and during COVID-19, the content of negative sentiments in Australia and the World News has increased compared to before the pandemic. On the contrary, in UK news, the proportion of negative sentiments during the pandemic has decreased.

4.7. The Guardian's Opinion section

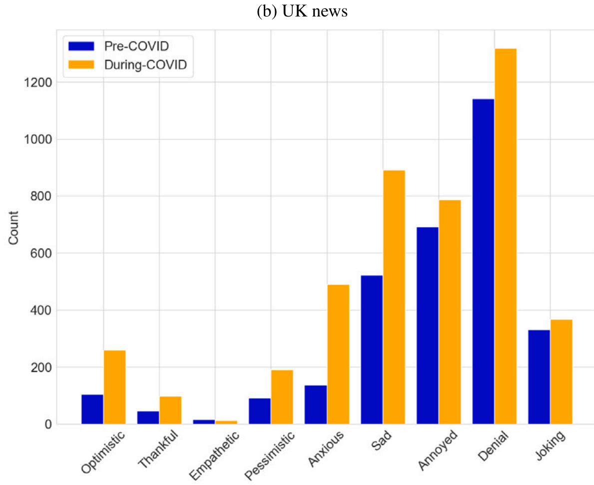
We transition to a focused examination of the sentiments from the Opinion section of The Guardian, presented in Fig. 13. We reintegrate the "official report" sentiment which was previously omitted from earlier analyses to focus on actual sentiments. As mentioned earlier, the "official report" category typically dominates due to the expected neutrality in news reporting. However, the Opinion section deviates from this trend, with "annoyed" and "denial" being the leading sentiments instead of "official report". This distinctive distribution aligns with the nature of opinion pieces, which often serve as a platform for expressing dissent and subjective viewpoints. The analysis of the Opinion section across the pre-pandemic and during-pandemic periods reveals surprising shifts in sentiment. We find that the "annoyed" and "denial" sentiments exhibit a decrease during the pandemic, while "anxious" and "sad" sentiments show a justifiable increase, reflecting the global distress caused by the crisis. Concurrently, the increase in "official report" sentiments during the pandemic is also natural, as opinion writers often refer to official statements and statistics to ground their arguments amidst the unfolding events. Furthermore, the "joking" sentiment has decreased suggesting a shift towards more serious discourse in opinion pieces during the pandemic, indicating the gravity of the situation and possibly a change in the role of humour in public commentary. Although these shifts are noticeable, they are not significant, which indicates that while the emotional tone of opinion pieces has shifted to accommodate the realities of the pandemic, the fundamental nature of the opinion section as a space for personal expression remained intact. It also suggests that opinion writers and readers have maintained emotional consistency, despite the transformative global events.



(a) Australia news



(b) UK news



(c) World news

Fig. 12. Sentiments detected in the Australia, UK and World News sections, before and during COVID-19.

4.8. *Sentiment-based trigram analysis*

We have reviewed the sentiment trend pre and during COVID-19, but this does not give an indication of why such sentiments were expressed. A way to know why sentiments such as "denial" or "annoyed" were expressed would be by reviewing the text associated with such sentiments and providing trigram analysis. Hence, this is

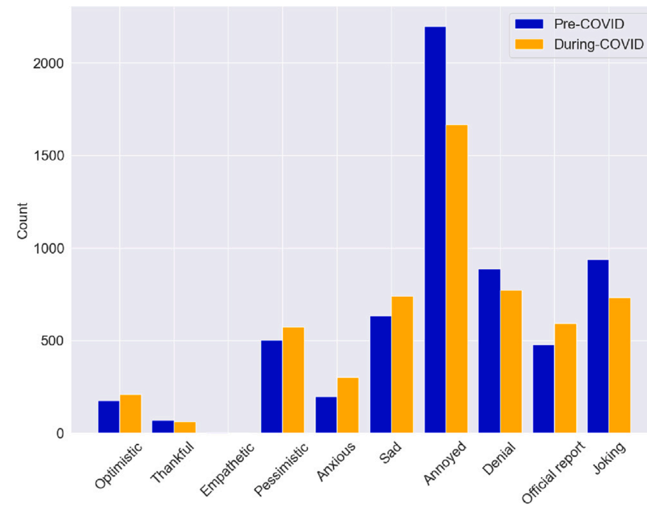


Fig. 13. Sentiments detected in the Opinion section before and during COVID-19.

moving backward, where we look at sentiments detected and then analyse the text further for those sentiments. In Fig. 2(a), a minor peak in the number of death cases in Australia during the early stages of COVID-19 is evident, occurring in the third quarter of 2020. Guided by the prevalent emotions identified in Fig. A.19(c) during the same period, which notably included "sad", "annoyed", and "denial", we selected the Australia news section and extracted trigrams. Fig. 14 presents phrases such as "health human service" and "hotel quarantine program" within these trigrams. "Health human service" likely pertains to medical services during the COVID-19 crisis, while "hotel quarantine program" likely relates to isolation policies implemented during the epidemic.

According to Fig. 2(a), after a small peak in the third quarter 2020, Australia's death toll stabilised until the third quarter 2021, hence we present Fig. 15 using the same method, which also contains "sad", "annoyed" and "denial"; we notice that there are no COVID-19 related trigrams except "local health district" and "chief health officer". This may indicate that media and public attention may have shifted to other events, issues or topics, rather than focusing on coverage and discussion of the epidemic. Moreover, this can also indicate the bias in news reporting by The Guardian that gets more attention from grim news articles.

As the number of deaths continued to rise, the public attention has once again shifted towards COVID-19. Notably, in Fig. 16, the reappearance of the trigrams "rapid antigen test" and "positive rapid antigen" suggests an increase in positive rapid antigen test results. This uptick reflects the escalating number of COVID-19 cases and the accelerated spread of the virus, signalling a renewed focus on the epidemic.

We can see from Fig. 17(a) that before COVID-19, the World News section focused on more diverse articles, ranging from human rights issues to the Prime Minister. Fig. 17(b) illustrates the trigrams extracted from The Guardian articles in the World News section during COVID-19. The analysis focused on filtering out sentiments associated with "denial", "annoyed", "anxious", and "sad" from 1st January 2020 to 31st March 2022. Moreover, Fig. 17(b), "health social care", "personal protective equipment" and "intensive care unit (ICU)" were frequently mentioned, reflecting the attention and importance paid to COVID-19 by the World News section, and also demonstrates the concentrated response to the challenges of the epidemic.

4.9. *Examples of sentiments detected*

We selected random samples of Australian news in Table 5, which include their predicted sentiments. We can see that in this table, the

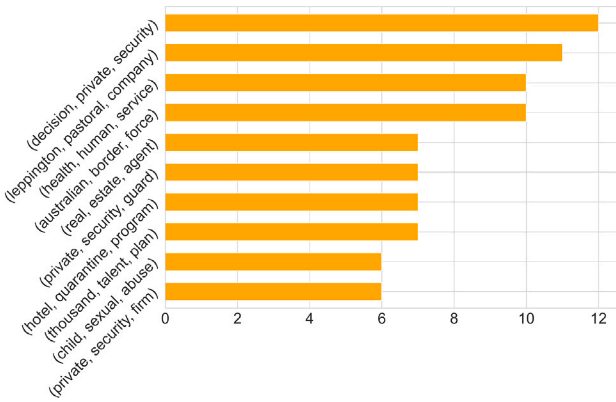


Fig. 14. Top 10 trigrams of the Australia news during 2020 third quarter (July–September) for sentiments "sad, annoyed and denial".

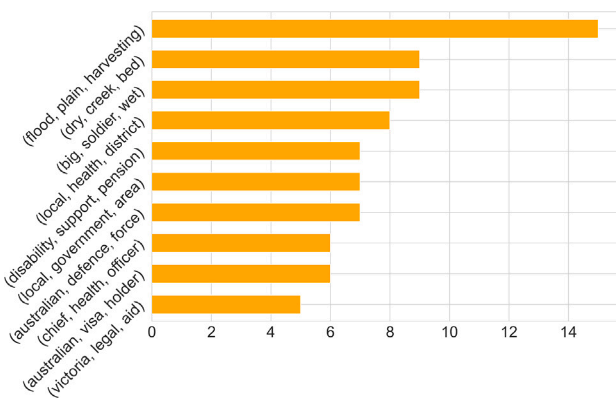


Fig. 15. Top 10 trigrams of the Australia news during 2021 third quarter (July–September) for sentiments "sad, annoyed and denial".

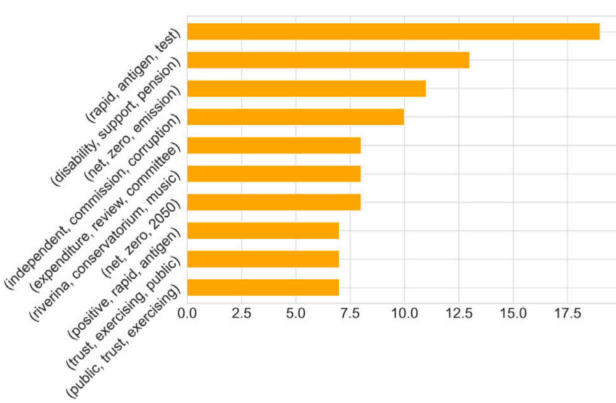
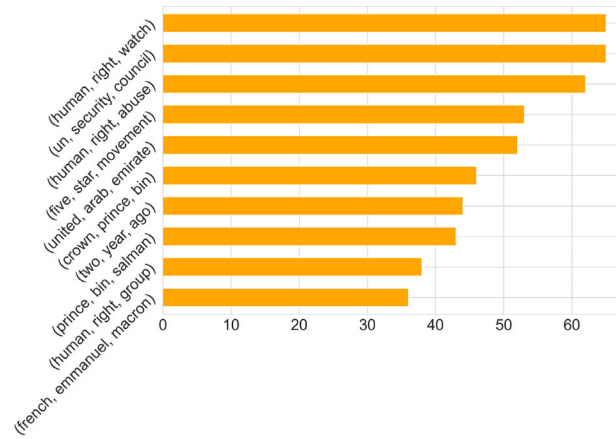
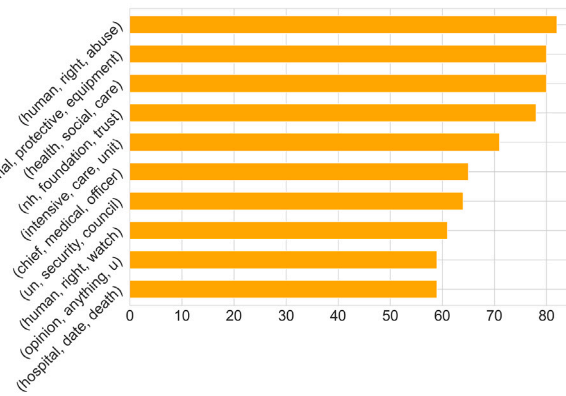


Fig. 16. Top 10 trigrams of the Australia news during 2021 fourth quarter (October–December).



(a) Before COVID-19



(b) During COVID-19

Fig. 17. Top 10 trigrams of the world news before and during the COVID-19 pandemic.

classification of "denial" and "joking" for the first article is incorrect. There is no indication of denial of reality or humorous elements in the text. Rather, the text discusses the restart of the Titanic II project by Clive Palmer, which does not involve denying reality or making jokes. A more appropriate classification for it would be "optimistic" as it portrays the restart of the project in a positive light, demonstrating hope and positivity for its future success.

The second row (dated 2021-10-31) in Table 5 describes the experiences of Adnan Choopani and his cousin Mehdi being detained in Australia's immigration detention system for up to eight years with suicide attempts. As for pessimism, the text portrays the various hardships and injustices suffered by Adnan and Mehdi in detention, which may evoke a pessimistic outlook on the future, especially considering their prolonged detention despite having lodged protection claims. The third row indicates anxiety among some unemployed individuals regarding the impending reduction in welfare benefits. They may feel apprehensive and fearful because this change could impact their livelihoods and quality of life.

We show further examples of sentiments detected in selected paragraphs of The Guardian newspaper articles from the UK and World news sections. We find that the tone of the first row in Table 6 is mainly optimistic, as it tells of a wonderful family holiday experience in a South African game reserve. The author recalls happy and beautiful times spent in nature and emphasises the positivity of the memories created during that time. The sadness may come from mentioning the death of Sudan, the last male northern white rhino. In the second row, the focus is on providing a historical account of royal tours, highlighting their significance in monarchial life throughout history. Therefore, the tone aligns more closely with an official report rather than expressing annoyance and joking sentiments. The third row describes the annoyance and denial of some of the parliamentarians over the new lockdown rules. They believe that the implementation of lockdown measures in low-risk areas is unreasonable, and expressed dissatisfaction and criticism about the lockdowns.

In Table 7, we find that the article discusses Greek scholars' potential pride in the widespread use of the Greek word "pandemic" globally, which may be interpreted as a joke or irony due to the severity of the pandemic with the seemingly pleasant context of using the word. Although the text does not directly express anger or annoyance, it suggests that Greek scholars are concerned about the detrimental

Table 5
Sentiment prediction outcomes for randomly selected article samples from Australia sections.

Sentiments	Sample articles	Public date	Section name
Denial, Joking	If the Titanic was the ship that was meant to be unsinkable, the Titanic II is the idea that seems to be un-killable. Six years after Australian mining magnate Clive Palmer declared he was going to build a replica of the Titanic, and three years after he had to suspend work on the project due to money troubles, Palmer has announced that work on the project will start up again. Building the boat – which will have the same interiors and cabin layout as the original vessel, complete with ballroom and Turkish baths, with reports passengers will be given period costumes to wear – has been a long-held dream of Palmer, an Australian businessman and conservative politician...	2018-10-24[a]	Australia news
Pessimistic, Sad	"Time can bring you down," Adnan Choopani sings, his words echoing off the walls of the detention centre compound, "time can bend your knees". Time is something Adnan, and his cousin Mehdi, know only too well. For eight years they have been held by Australia's immigration detention regime, offshore and on. They have watched friends burn themselves to death and known the despair that has led them to attempt suicide themselves. They have been beaten and abused, jailed without reason. They have grown from boys into men in that time. Fifteen and 16 when they arrived in Australia seeking sanctuary, they are now 23. Despite their claims for protection being formally recognised more than half a decade ago, they remain in detention...	2021-10-31[b]	Australia news
Anxious	Jobseekers in locked-down Melbourne are bracing for a "devastating" $300 cut to welfare benefits that will hit only two weeks after stage four restrictions are expected to end. The federal government's plan to taper the coronavirus supplement on 25 September is expected to reduce the incomes of about 2.3 million unemployed people, single parents and students across Australia. But jobseekers and their advocates expect the economic cliff to hit Melbourne particularly hard, with the cut coming only 12 days after the scheduled end to restrictions that shut key industries and imposed a nightly curfew. Cassandra Francisco, 23, who is living under stage 4 restrictions in Footscray in Melbourne's west, said she was awaiting the change with "absolute dread"."I'm just going to barely be able to cover my rent," she said. "I'm already stressed by the situation."...	2020-8-30[c]	Australia news

[a] https://www.theguardian.com/australia-news/2018/oct/24/titanic-2-australian-billionaire-refloats-dream-clive-palmer
[b] https://www.theguardian.com/australia-news/2021/nov/01/time-can-break-your-heart-the-harsh-toll-of-eight-years-in-australian-immigration-detention
[c] https://www.theguardian.com/australia-news/2020/aug/31/jobseekers-in-locked-down-melbourne-brace-for-devastating-300-cut-to-welfare

Table 6
Sentiment prediction outcomes for randomly selected article samples from UK sections.

Sentiments	Sample articles	Public date	Section name
Optimistic, Sad	Monday One of the best family holidays we ever had was on a game reserve in South Africa about 10 years ago. Back then it was normally tricky to get the kids out of bed much before 11, but there were no moans about getting up at six every morning to go on an escorted drive to watch the animals. There was something so magical about being in the presence of such natural beauty. The highlight was coming across three rhinos among a clump of trees. Our jeep crept up to within about 20 metres and we all sat in silence for the best part of an hour, overcome with wonder. So the death of Sudan, the last male northern white rhino, felt more personal than it otherwise might have done...	2018-03-23[a]	UK news
Annoyed, Joking	Royal tours have long been a central feature of monarchical life. It is what they do. As the Queen says: "We have to be seen to be believed." Medieval monarchs toured their realms obsessively in order to show they were still alive. It also helped keep their populations in order and allowed them to display their magnificence and power. Henry II's legs grew bandy as he rode continuously across France, England and Ireland in the 12th century. Elizabeth I's tours, 400 years later, wended their way round the country: she spoke to ordinary folk encountered en route and accepted gifts from the burghers of the towns that she and her 300-wagon baggage train passed through...	2022-03-29[b]	UK news
Annoyed, Denial	Residents, Tory MPs and police have rounded on the government's handling of new lockdown rules for northern England, while Muslim leaders raised concerns that communities were being scapegoated. On a day of confusion and anger over measures affecting 4.6 million people, police federations warned that new laws barring visitors from private homes or gardens on the eve of Eid may be impossible to enforce. Some MPs expressed anger at the measures in areas with low coronavirus cases, and there was criticism of the announcement being made on Twitter at 9.16pm on Thursday, less than three hours before the rules were imposed. Boris Johnson refused to condemn a fellow Tory MP for saying that Muslims were "just not taking the pandemic seriously"...	2020-07-31[c]	UK news

[a] https://www.theguardian.com/uk-news/2018/mar/23/rhino-death-rotten-kipper-farage-national-humiliation-brexit?CMP=share_btn_url
[b] https://www.theguardian.com/uk-news/2022/mar/29/we-have-to-be-seen-to-be-believed-the-endurance-of-the-royal-tour?CMP=share_btn_url
[c] https://www.theguardian.com/uk-news/2020/jul/31/matt-hancock-defends-last-minute-northern-coronavirus-lockdown?CMP=share_btn_url

Table 7
Sentiment prediction outcomes for randomly selected article samples from World news sections.

Sentiments	Sample articles	Public date	Section name
Annoyed, Joking	Usually, Professor Georgios Babiniotis would take pride in the fact that the Greek word "pandemic" – previously hardly ever uttered – had become the word on everyone's lips. After all, the term that conjures the scourge of our times offers cast-iron proof of the legacy of Europe's oldest language. Wholly Greek in derivation – pan means all, demos means people – its usage shot up by more than 57,000% last year according to Oxford English Dictionary lexicographers. But these days, Greece's foremost linguist is less mindful of how the language has enriched global vocabulary, and more concerned about the corrosive effects of coronavirus closer to home...	2021-01-31[a]	World news
Annoyed, Denial	Simon Tisdall (Trump's peace plan is a gross betrayal of Afghanistan, Journal, 20 August) is completely correct in his analysis of the US peace initiative in Afghanistan: it is all about enabling the withdrawal of US troops and little about the future of the Afghan people; "details", such as the shape of the future government of the country, are, apparently, to be settled later. This week saw one historic occasion for Afghanistan, the 100th anniversary of its independence from Britain, but the deal being negotiated in Doha more closely resembles the end of the Soviet engagement in 1989. Desperate to stop a war they were losing, the Soviets negotiated the end of combat operations in 1989 and withdrew the bulk of their troops. Three years later, under pressure from the west, "cooperation troops" withdrew...	2019-08-23[b]	World news
Annoyed, Denial	Labour has accused the government of "monumental mistakes" in its handling of the pandemic, as ministers continued to insist they did everything they could to try prevent more than 100,000 deaths from the virus. The shadow health secretary, Jonathan Ashworth, said a "litany of errors" had led to the UK having the fifth-highest death toll in the world and the highest death rate relative to its population. "I just do not believe that the government did do everything we could," he told the BBC Radio 4's Today programme. "We all accept these are challenging times for any government. This is a virus which has swept across the world with speed and severity and it continues to spread ferociously ... But monumental mistakes have been made...	2021-01-27[c]	World news

[a] https://www.theguardian.com/world/2021/jan/31/the-greeks-had-a-word-for-it-until-now-as-language-is-deluged-by-english-terms?CMP=share_btn_url

[b] https://www.theguardian.com/world/2019/aug/23/afghanistan-at-risk-of-being-abandoned?CMP=share_btn_url

[c] https://www.theguardian.com/world/2021/jan/27/monumental-mistakes-made-over-handling-of-covid-says-labour?CMP=share_btn_url

impact of the epidemic on their homeland, which could manifest as mild annoyance or dissatisfaction. In the second row of Table 7, the author expresses annoyance with the United States peace initiative, suggesting that it merely serves as a means to facilitate the withdrawal of their troops without genuinely considering the future and interests of the Afghan people. This dissatisfaction reflects a critical perspective on policy decisions, questioning the sincerity and effectiveness of the initiative. It can be interpreted as a sense of annoyance towards the lack of genuine consideration for the Afghan people's future. Additionally, the author's stance implies a denial of the effectiveness or goodwill of the peace initiative, highlighting scepticism towards its true intentions and outcomes. In the third row of Table 7, the Labour Party (opposition) accused the government of making a "significant mistake" and expressed annoyance, believing that the government's actions were not decisive or effective enough. They pointed out that the United Kingdom had the fifth-highest death toll in the world, with the highest mortality rate relative to the population. It also reflects a denial of these measures.

5. Discussion

In our results, it is evident that the news articles selected from The Guardian during the COVID-19 pandemic mostly contain negative sentiments (denial, sad, annoyed and anxious). In a related study, Chandra and Krishna [10] reviewed tweets from specific regions in India, such as Maharashtra and Delhi, during the surge of COVID-19 cases. The results revealed a spectrum of sentiments, predominantly optimism, annoyance, and humour (joking), which were vividly expressed through X (Twitter) during the pandemic where the sentiments were predominantly positive, but it constitutes a very small proportion in our results. This discrepancy between news media in our study and social media (X) in the past study highlights the differing roles that various media types play in public sentiment during crises. It suggests that social media might offer a more diversified emotional reflection, compared to the often grim narrative found in traditional news media.

The differences in emotional expression between social media and traditional news media are pivotal, especially when considering their impact on public perception and behaviour. According to Anspach and Carlson [87], individuals are more likely to trust and be influenced by the information conveyed through social media comments, even when it deviates from their pre-existing beliefs or is factually incorrect. This tendency to trust social media can be attributed to the perceived immediacy and personal connection users feel towards content creators, which is not as prevalent in traditional media. Anspach and Carlson [87] reported that despite a general distrust in the news shared through social media, individuals were paradoxically more inclined to believe the information provided by social media posts over content that has been professionally reviewed. This phenomenon can lead to significant implications for public health, where accurate information dissemination is crucial. Our study indicated that denial is the most prevalent sentiment, followed by sadness and annoyance. Therefore, it is apparent that the pervasive negative sentiments in COVID-19 news media could have exacerbated a despondent public mood, contrasting with the emotional variety observed on social media. Based on these findings, it is crucial to improve the transparency and accuracy of information dissemination during pandemics to manage public sentiment effectively. Furthermore, we note that such negative sentiments were also prevalent in The Guardian prior to COVID-19 (Figs. 11, 12 and 13). Furthermore, Chandra and Krishna reported that positive sentiments such as "optimistic" and "joking" were more prominent in tweets (see Fig. 18). However, in our study (Fig. 5), apart from official reports, we found that negative sentiments such as "annoyed" and "denial" were the most prominent. The difference may be due to the style of expression among different media sources (i.e. Twitter vs The Guardian). On social media platforms, everyone could post tweets to share their thoughts about COVID-19, covering sentiments from all users. In traditional media such as The Guardian, news reports are written by journalists and reviewed by newspaper editors who typically

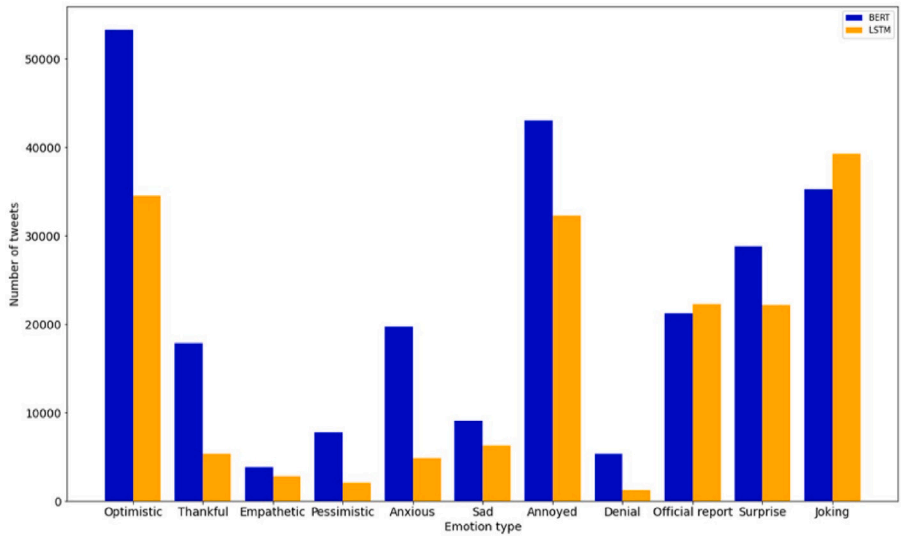


Fig. 18. Distribution of sentiments predicted for the Indian tweet dataset by the LSTM and BERT models taken from Chandra and Krishna (2021) [10].

employ a formal tone to describe various events.

Over the past few decades, with the Internet and social media, newspapers have been challenged to sustain themselves and update their business models with paywalls [88]. Accountability, transparency and factual news reporting are continuously being demanded from the government and the public [89]. Sensationalised news is becoming more common since traditional news media needs to compete with social media platforms and influencers, and it is a common practice even to cover news about social media *influencers* who are taking over as celebrities [90]. The bias in newspaper reporting which leans towards a grim narrative and political narrative is well known [91–94]. In the case of COVID-19, Carney et al. [95] presented a study on media biases, and political narratives in the major UK newspapers and reported that progressive and conservative newspapers reflected age-based stereotypes and paternalism towards older people. We note that different newspapers have their own political biases [96] as some have left and right-wing political leanings and The Guardian is well known as a centre-left-leaning newspaper [97,98]. Although we reported a grim narrative from The Guardian pre and during COVID-19 through sentiment analysis, such a style of reporting by newspapers is common. It is typically used to gain attention from the public to increase or maintain newspaper sales [88].

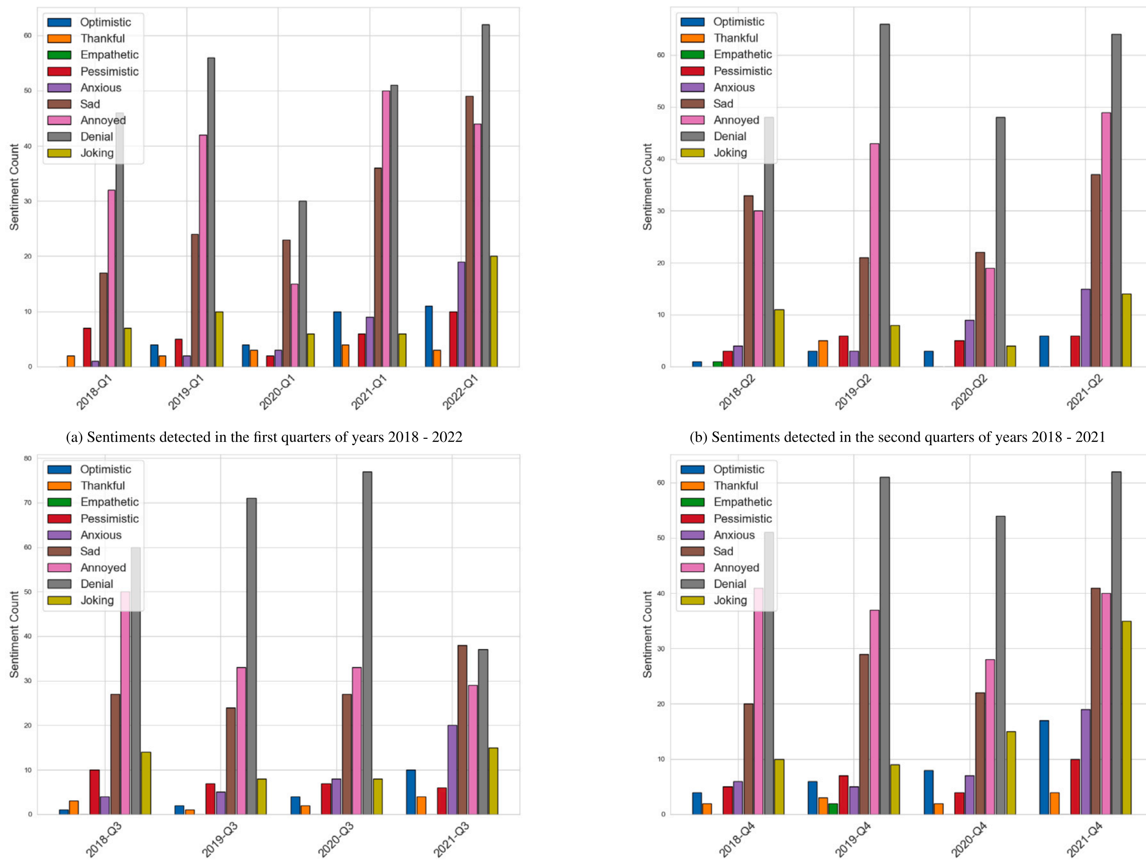
A limitation of this study is that we fine-tuned the model using the SenWave dataset, which consists of millions of tweets and Weibo messages sourced from social media. The information posted on social media often contains numerous abbreviations, internet slang and emojis, which posed a challenge in training our model for sentiment analysis. Furthermore, when training and testing the model, we utilise the X (Tweeter) dataset, which typically features shorter text lengths when compared to The Guardian articles. As a result, it may fail to capture the same richness and complexity of language features, potentially leading to performance degradation when processing longer texts. Furthermore, we chose The Guardian newspaper for the case study, there is an open dataset for it on Kaggle. Therefore, our findings are limited to regions, particularly Australia and the UK, where The Guardian's readership is more prominent. Since the news articles were exclusively sourced from The Guardian, the sentiment analyses may not reflect the global sentiment towards the COVID-19 pandemic. Our study gives a glimpse of how a particular Western newspaper reported COVID-19 news events, not just for their key region of interest but for the global events as given in their international news reporting. Our framework is general and other newspapers can be considered in future works. Additionally, with the increasing prevalence of social media platforms such as Twitter, Instagram, Facebook and Weibo, people are more inclined to share their thoughts and feelings in real time. Information obtained from news media primarily represents the editorial stance of the outlet, rather than the authentic sentiments of the populace towards the COVID-19 pandemic.

In future work, we can utilise our framework to conduct sentiment analysis on news reports from various news media outlets during the COVID-19 pandemic. We can review public sentiments in different countries and gain a more comprehensive understanding of the global COVID-19 pandemic. Novel LLMs can be employed to conduct sentiment analysis utilising the framework in this study on news related to epidemics and major events such as wars and political unrest, including the Israel-Hamas and the Ukraine–Russian wars. Additionally, combining topic thematic modelling with sentiment analysis could provide deeper insights into emerging topics relevant to government policies, such as lockdowns and vaccination programs. Furthermore, advanced LLMs can be utilised, such as the Generative Pre-trained Transformer (GPT) models [99] (including GPT-4o, Grok and Gemini) for related NLP tasks such as review of abusive comments, translation of local languages for sentiment analysis, and modelling major topics during COVID-19. Furthermore, future research in sentiment analysis can utilise novel methodologies such as multimodal deep learning models, integrating various media forms including text, audio, and video. Specifically, integrating multimodal NLP models into our framework can facilitate the emotional analysis of users' textual, vocal, and facial inputs. This can further be enhanced to support and monitor an individual's mental health status on social media, and potentially detect signs of self-harm and suicidal tendencies.

6. Conclusion

During the COVID-19 pandemic, various media platforms engaged in extensive news reporting, reflecting the complex emotional fluctuations of the public. We employed large language models, specifically BERT and RoBERTa for sentiment analysis on news and opinion articles from The Guardian. The results show that both models have similar capabilities when predicting refined sentiments from the SenWave dataset.

Our results indicated that in the initial stages of the pandemic, public sentiment was primarily focused on the urgent response to the crisis. As the situation evolved, the emotional emphasis shifted towards addressing the enduring impacts on health and the economy. During the COVID-19 pandemic, the sentiments detected in news articles indicated a significant increase in the frequency of articles with either one



(a) Sentiments detected in the first quarters of years 2018 - 2022

(b) Sentiments detected in the second quarters of years 2018 - 2021

(c) Sentiments detected in the third quarters of years 2018 - 2021

(d) Sentiments detected in the fourth quarters of years 2018 - 2021

Fig. A.19. Quarterly sentiment counts in Australia news section of The Guardian using the RoBERTa model.

or two emotional labels, highlighting an increased focus on dominant sentiments in news media reporting. We found that Australian news articles focused more on addressing the immediate crisis, while the UK reported greater emphasis on social impacts and mental health. This could be attributed to the UK experiencing a higher severity during the pandemic, which is reflected in the prevalence of negative sentiments in its news reporting, although the reports also frequently exhibited a positive perspective. We also found a discrepancy between our study of news media and an earlier study, where social media reported a more diverse emotional expression during COVID-19. Finally, our study highlighted the grim narrative used as a style of expression in The Guardian as traditional news media with overall dominance of negative sentiments, pre and during COVID-19 across the respective news sections (including Australia, UK, World News, and Opinion).

Code and data availability

We provide open source code and data that can be used to extend this study [8]. Given in Github repo.

CRediT authorship contribution statement

Rohitash Chandra: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Baicheng Zhu:** Writing – original draft, Software, Methodology, Data curation. **Qingying Fang:** Writing – original draft, Software, Methodology, Data curation. **Eka Shinjikashvili:** Writing – review & editing, Validation, Supervision, Investigation.

[8] https://github.com/sydney-machine-learning/sentimentanalysis-COVID19news

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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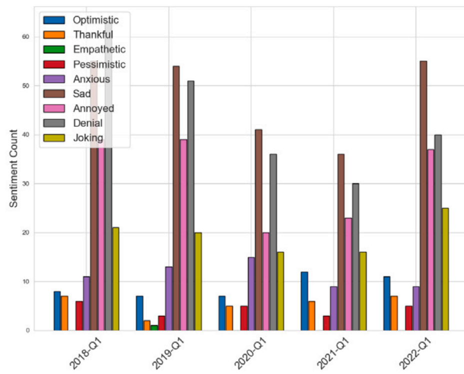
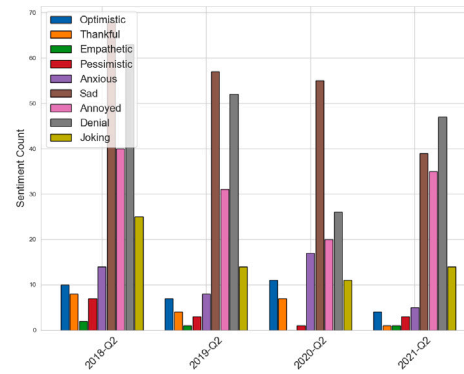
Appendix. Quarterly analysis

In Fig. A.19, which provides the quarterly sentiment distribution in Australia, we focus on the first quarter of 2022. The RoBERTa model highlights "sad" as the sentiment with the most significant increase, aligning with earlier observations that RoBERTa is more attuned to identifying sadness. In the case of 2021-Q4 in Fig. A.19(d), the model predictions are as expected, with fewer instances of "sad" compared to 2022-Q1. The overall trend is clear as there is a general increase in negative sentiment from pre-pandemic to the pandemic period, underscoring the emotional toll of the COVID-19 crisis as reflected in media coverage across the quarters.

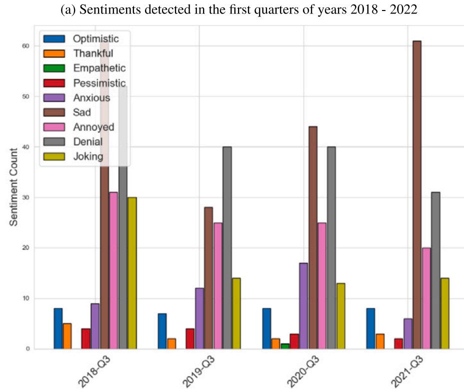
Figs. A.20, representing the quarterly sentiment distribution in the UK, contrast with the patterns observed for Australia. According to insights from Fig. 2(b), the UK's death toll due to COVID-19 exceeded 60,000 in the first quarter of 2021. Interestingly, during this quarter, most of the negative sentiments such as "sad" and "anxious" decreased compared to the pre-pandemic period, while positive emotions such as "optimistic" and "thankful" saw a slight increase, especially as identified by the RoBERTa model. Similarly, despite the death toll



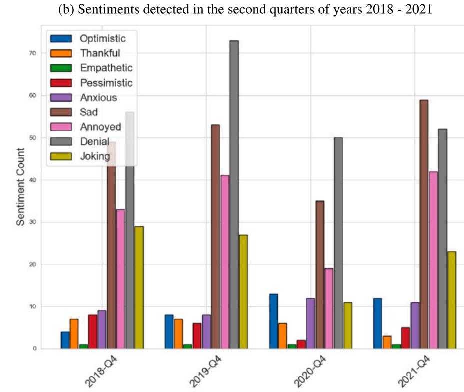
(a) Sentiments detected in the first quarters of years 2018 - 2022



(b) Sentiments detected in the second quarters of years 2018 - 2021

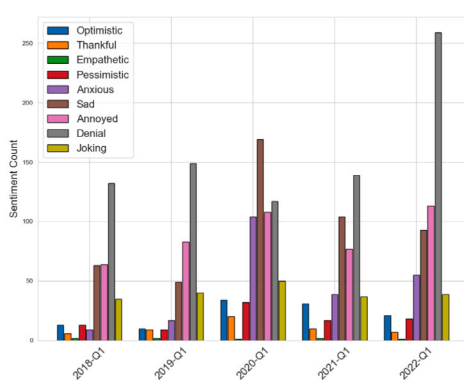


(c) Sentiments detected in the third quarters of years 2018 - 2021

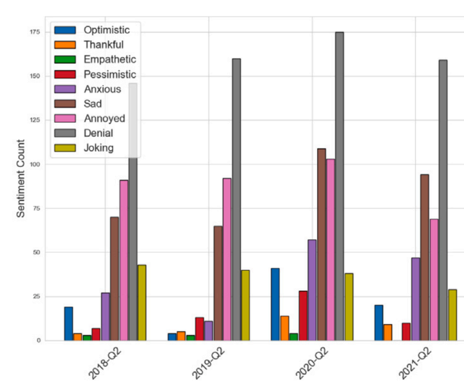


(d) Sentiments detected in the fourth quarters of years 2018 - 2021

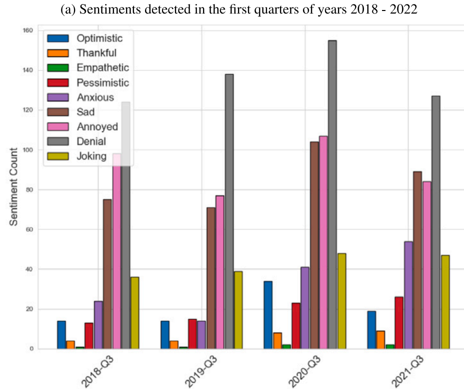
Fig. A.20. Quarterly sentiment counts in the UK news section of The Guardian using the RoBERTa model.



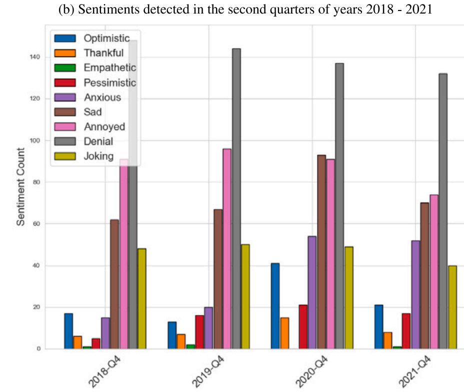
(a) Sentiments detected in the first quarters of years 2018 - 2022



(b) Sentiments detected in the second quarters of years 2018 - 2021



(c) Sentiments detected in the third quarters of years 2018 - 2021



(d) Sentiments detected in the fourth quarters of years 2018 - 2021

Fig. A.21. Quarterly sentiment counts in the world news section of The Guardian using the RoBERTa model.

reaching over 50,000 in the second quarter of 2020, there was no significant rise in negative emotions detected by the model. This finding is peculiar when juxtaposed with the Australian data, where a marked increase in negative sentiments correlated with the progression from pre-pandemic to pandemic periods. The sentiment distribution in the UK, as reflected in the news, seems to lack a consistent temporal pattern in response to the unfolding crisis.

Fig. A.21 presents the quarterly emotion distribution worldwide, where the model detected a higher number of negative sentiments such as “anxious” and “sad” throughout all four quarters of 2020 when compared to the pre-pandemic years.

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